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Failure analysis of ternary lithium-ion batteries throughout the entire life cycling at high temperature

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ABSTRACT

The operation life is a key factor affecting the cost and application of lithium-ion batteries. This article investigates the changes in discharge capacity, median voltage, and full charge DC internal resistance of the 25Ah ternary ($LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$/graphite) lithium-ion battery during full life cycles at 45 °C and 2000 cycles at 25 °C for comparison. The batteries before and after cycling were disassembled, and the structure, morphology, interface characteristics, element content of the cathode and anode plates of the battery were characterized. By analyzing the failure factors of the performance of the ternary batteries during the 45 °C cycling, a reaction mechanism for the rapid decline of high-temperature cycling performance of ternary batteries has been proposed. The results show that the performance degradation of the ternary lithium-ion batteries in the whole life operated at high temperature is characterized by slow decline in the initial stage and rapid drop in the latter stage. Further analysis of physical and chemical performance revealed irreversible damage to both the cathode and anode. The dissolution of transition metal ions from the cathode and their deposition on the anode surface catalyze the decomposition of electrolyte solvents to produce a large amount of gas. Gassing, accompanied by the deposition of Li_2CO_3 and the thickening of SEI, leads to an increase in the internal resistance of the battery. The coupling between gassing and increased internal resistance is responsible for the rapid drop in the performance of the ternary batteries during high-temperature cycling.

1. Introduction

In today's world, a significant portion of electricity generation and transportation relies on non-renewable fossil fuels, contributing to environmental challenges like global warming due to CO_2 emissions. To address this and secure a sustainable future, prioritizing renewable and clean energy sources such as solar, wind, and geothermal power is imperative. However, many renewable sources are influenced by factors like season and location, necessitating efficient energy storage solutions. Lithium-ion batteries (LIBs) stand out as cutting-edge technology for various energy storage applications, boasting high energy density, power density, long cycle life, and minimal self-discharge [1,2].

In 2001, Ohzuku and Makimura [3] reported on $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$, also known as 333-type ternary cathode material. This material combines the advantages of three-layered materials: $LiCoO_2$, $LiNiO_2$, and $LiMnO_2$, demonstrating the synergistic effect of cobalt, nickel, and manganese elements. It offers high specific capacity, good cycling performance, and other benefits, quickly attracting the attention of researchers. To meet the demand for high energy density in power batteries, the positive material system of ternary batteries of nickel-cobalt-manganese oxides (NCM) has evolved from low-nickel content NCM_{333} ($LiNi_{1/3}Mn_{1/3}Co_{1/3}O_2$), NCM532 ($LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$), etc., towards high-nickel content NCM622 ($LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$), NCM811 ($LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$).

However, with the increase of nickel content, the safety performance and lifespan of batteries have significantly decreased [4]. Taking NCM622 ternary power batteries as an example, their cycle life is less than 2000 cycles, while the cycle life of lithium iron phosphate batteries is greater than 4000 cycles. There are many factors leading to the short life of ternary batteries, mainly including loss of active materials [5], electrolyte decomposition [6], changes in the crystal structure of cathode materials [7]. The main reason for the poor thermal performance of

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ternary batteries is the poor thermal stability of the cathode NCM materials. During the thermal runaway process of the battery, the decomposition of NCM will also release O_2, leading to a fast temperature rise rate and high harm when the ternary battery is in thermal runaway [8-10].

The key to solving the above problems is to understand the performance degradation patterns of ternary batteries throughout the entire lifespan of ternary batteries and the mechanisms behind rapid performance decline in the later stages. Recent studies on the failure mechanisms of NCM cathode batteries have highlighted several key factors contributing to their degradation. Investigations suggest that degradation is often accelerated by a combination of factors including transition metal dissolution, structural instability, electrolyte decomposition, and surface reactions at the electrode-electrolyte interface [11-14]. These studies have emphasized the importance of understanding the intricate interplay between these factors, as well as the role of operating conditions such as temperature and voltage cycling, in order to develop strategies for mitigating degradation and improving the longevity and performance of NCM cathode batteries.

There are limited studies focusing on the anode side as a cause of ternary battery failure. These studies investigated anode side causes for the failure of the ternary lithium batteries at a given temperature, e.g., at −10 °C [15] or 45 °C [16]. It was reported [15] that NCM ternary pouch battery show about 25% irreversible capacity after 500 cycles at −10 °C due to lithium plating on the surface of anode. Gas (mainly CO) is formed in the pouch battery when stored at room temperature after the low temperature cycles. The operation stability of high-lithium NCM ($LiNi_{0.83}Co_{0.12}Mn_{0.05}O_2$/graphite–$SiO_x$) pouch batteries at high temperature (45 °C) were studied by Wang et al [16]. The studied samples were mostly based on coin-cell batteries. The batteries exhibit 25% loss in capacity within 280 cycles due to deposition of transition metal ions from cathode on the anode and loss of lithium. This work did not investigate the formation of gas for the batteries during cycles at high temperatures.

This study investigated the cycling of commercial ternary batteries over their full life cycle at 45 °C, and over 2000 cycles at 25 °C for comparison. The degradation of battery performance and the deterioration of battery components formed a coupling relationship. The study was focused on the failure mechanism of metal element deposition on the anode surface and the generation of gas at the anode interface. The goal of this study on the failure mechanism was to provide guide for the safe operation and cascading utilization of ternary batteries and offer insights for improving ternary battery performance.

2. Experimental

In this work 25 Ah ternary batteries from a well-known power battery manufacturer were selected as research samples. The cathode material used was NCM532 ($LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$), and the anode material was graphite, with polypropylene separator and $LiPF_6$ in dimethyl carbonate (DMC), ethylene carbonate (EC) and propylene carbonate (PC) as electrolyte. The battery has a length of 227 ± 1.0 mm, a width of 161 ± 1.5 mm, and a thickness of 7.45 ± 0.16 mm. The nominal voltage of the battery is 3.65 V. The internal resistance of the battery at 30% state of charge (SOC) and 1 KHz frequency is 1.0 ± 0.15 mΩ.

A Neware BTS-5 V/50 A or BTS-5 V/100 A charge-discharge testing system was used to study the cycling performance of the 25 Ah ternary battery samples at 25 °C and 45 °C under 1C current. The charging cutoff voltage was set at 4.15 V, and the discharging cutoff voltage was set at 2.7 V. During discharge, lithium ions leave the anode and migrate back to the cathode. Lithium without electrochemical activity remains on the surface of the anode. Therefore, the characterization of lithium plating is better suited for the battery in a discharged state. The battery was first discharged to 0% SOC, and then it was disassembled in an argon atmosphere glove box (with water and oxygen content both less than 0.1 ppm). The cathode, anode, and separator were washed with dimethyl carbonate and dried overnight in the glove box. Finally, the samples were transferred using a sealed box for subsequent physicochemical analysis.

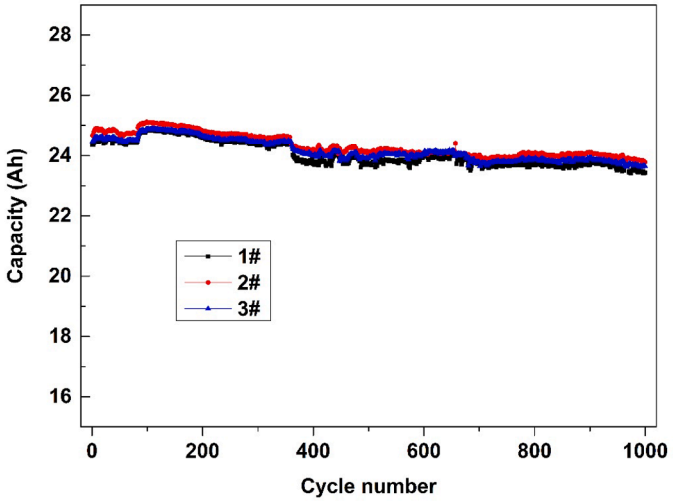


Fig. 1. Performance curves of three randomly selected batteries cycled at 25 °C with a 1C current.

The surface morphology of the positive and anodes under different conditions was observed using scanning electron microscopy (SEM, FEI Nova Nano SEM 450). The crystal structure of the cathode materials was analyzed using X-ray diffraction (XRD) with a German Bruker instrument (D8, Bruker), which used copper target material (λ= 0.154 nm) for the analysis of crystal structure changes in the cathode material.

During the later stages of cycling at 45 °C, the battery began to produce gas. After 1000 cycles, a rubber pad was sealed with adhesive on the surface of the battery's aluminum-plastic film. An Agilent airtight needle was used to penetrate the rubber pad and the aluminum-plastic film to extract gas samples. Since the rubber pad was adhered to the battery's surface with sealant, the airtight needle maintained a sealed environment inside the battery during and after the sampling process. The gas composition and content of battery gas samples were analyzed using an Agilent 7890B gas chromatograph. A 0.5 mL gas sample was taken and separated using a Haye Sep DB column. The analysis was performed with two detectors: a thermal conductivity detector (TCD) and a flame ionization detector (FID), with argon as the carrier gas at a flow rate of 38 cc/min. The anode material was analyzed for element content using a Thermo ICAP6000 atomic emission spectrometer.

Microstructure and chemical environment analysis of the anode interface for the battery were analyzed after the cycling test. In this experiment, transmission electron microscopy (TEM, FEI Tecnai G2 F30 S-TWIN), infrared spectroscopy (IR, BRUKER ALPHA II), and X-ray photoelectron spectroscopy (XPS) with an ESCALAB 250Xi (Thermo) were used to analyze the anode of the ternary battery samples. The aim was to study the changes in the SEI and crystal structure at the anode interface before and after battery failure.

3. Results and discussion

3.1. Study of electrochemical performance failure at room temperature and high temperature

The consistency of battery performance serves as a reliable foundation for this experimental data. Consistent battery performance helps reveal differences in performance under varying conditions, rather than simply reflect variations in the manufacturing process. To evaluate battery consistency, three batteries were randomly selected from a batch of 50 and subjected to 1000 cycles at 25 °C with a 1C current to assess their performance, as shown in Fig. 1. The initial discharge capacities of

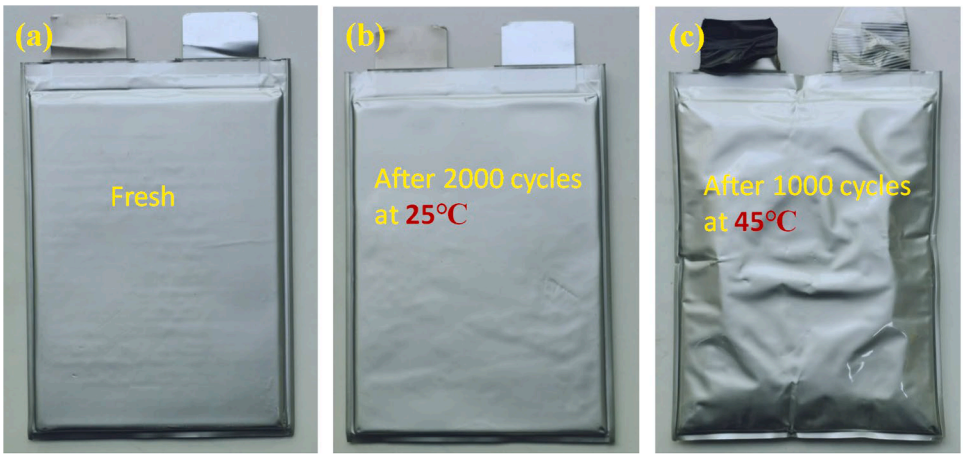


Fig. 2. Images of the pouch batteries at different states: (a) pristine battery, (b) battery after 2000 cycles at 25 °C, and (c) battery after 1000 cycles at 45 °C.

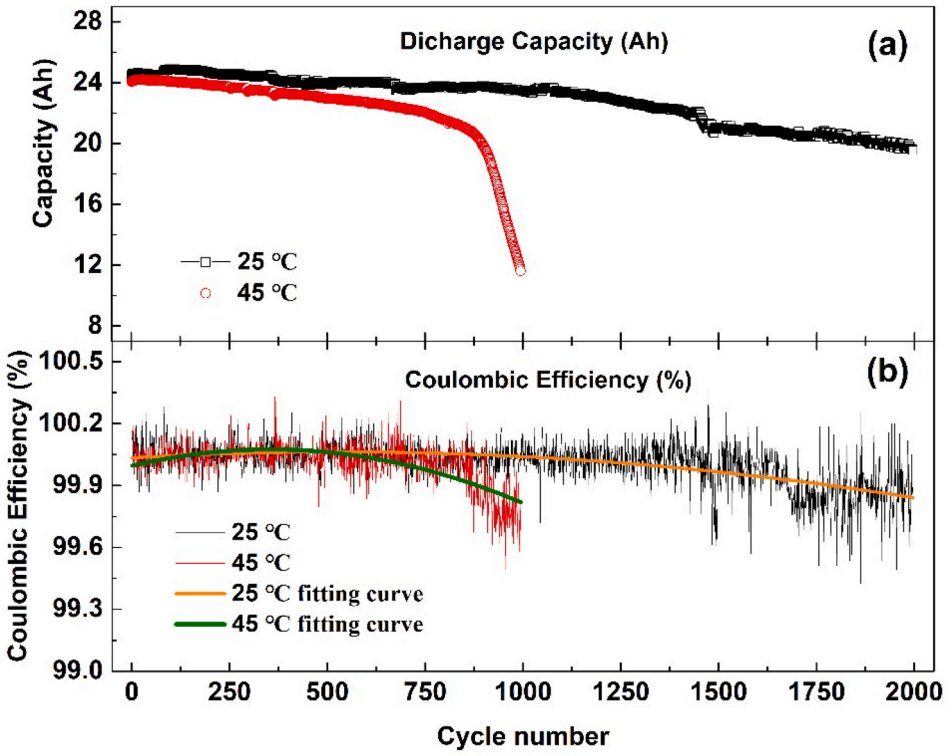


Fig. 3. Discharge capacity (a) and coulombic efficiency (b) versus the cycle number at 45 °C and 25 °C (the voltage range of charge - discharge is 2.7 V - 4.15 V).

the three batteries were 24.3 Ah, 24.7 Ah, and 24.6 Ah, respectively, with a capacity deviation of less than 0.2%. After 1000 cycles, the discharge capacities were 23.4 Ah, 23.8 Ah, and 23.6 Ah, with the deviation still within 0.2%. This indicates that the batteries exhibit good consistency and meet the requirements of the experiment.

The electrochemical performance of a lithium-ion battery is significantly influenced by temperature, where high temperatures accelerate lithium-ion migration, reducing internal resistance. However, high temperatures also accelerate battery aging, even leading to rapid degradation of battery performance. Fig. 2 shows the appearance of pristine battery, and that after 1000 cycles at two different temperatures. It can be observed that the battery sample cycled at 25 °C for 2000 cycles shows no significant change compared to the pristine battery. However, the battery cycled at 45 °C for 1000 cycles exhibits severe gas generation, and the battery was completely deformed. To study the high-temperature failure mechanism of ternary batteries, battery discharge capacity, coulombic efficiency, charge-discharge curves, midpoint voltage, discharge energy, and DC internal resistance of the batteries operated at 45 °C were measured and the results are compared with the performance data of the same batteries operated at 25 °C.

3.1.1. *Discharge capacity and Coulombic efficiency*

Fig. 3 shows discharge capacity and coulombic efficiency versus cycles for the battery operated at the two temperatures. During cycling at 25 °C, the discharge capacity gradually decreased, from the initial 24.5 Ah to 19.5 Ah after 2000 cycles, with a capacity drop by 20%. The discharge capacity after 1000 cycles was 23.4 Ah, a 5% drop. However, during cycles 1000-2000, the capacity dropped by 15%, indicating an accelerated performance degradation in the later stages of life. When the battery is cycled at 45 °C, the entire life degradation can be divided into two stages: a slow drop in the first 800 cycles (only 2.5% drop) and a rapid drop during 800–2000 cycles (49% drop). The performance

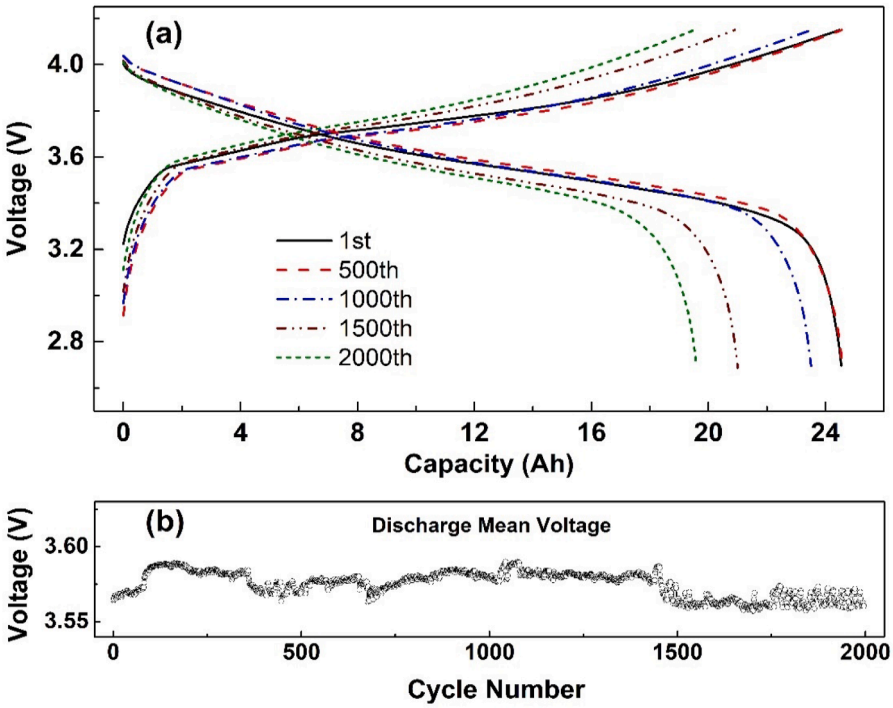


Fig. 4. Voltage-capacity curves (a) and variation of midpoint voltage with cycle number (b) of the ternary battery cycling at 25 °C.

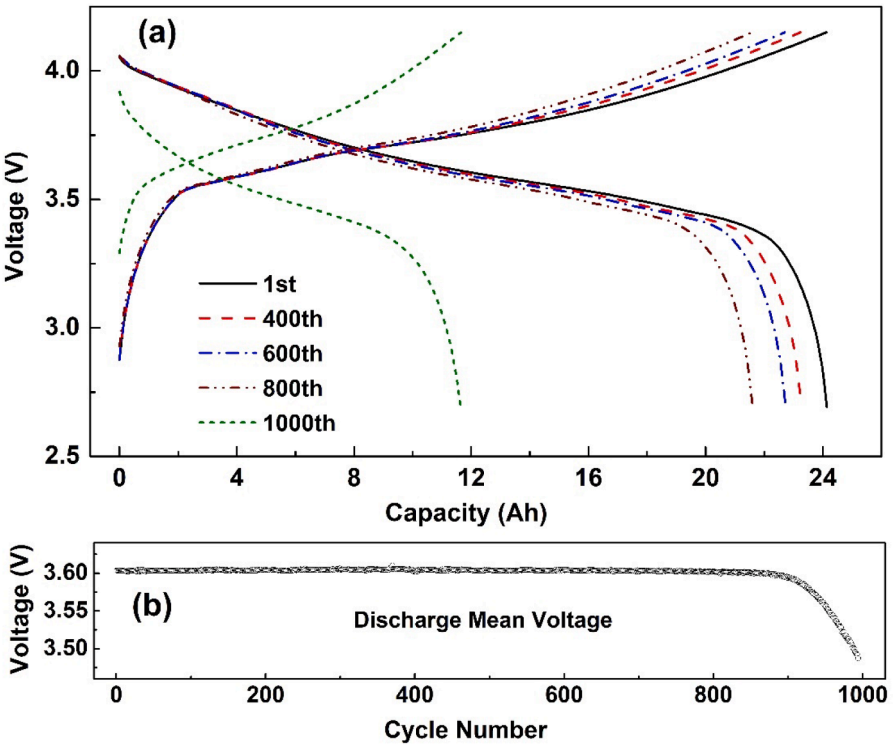


Fig. 5. Voltage-capacity curves (a) and variation of midpoint voltage with cycle number (b) of the ternary battery cycling at 45 °C.

degradation rate is significantly higher after 800 cycles when operated at 45 °C.

From Fig. 3(b), it can be seen that with an increase in the number of cycles, the coulombic efficiency decreased, especially during the later stages of cycling at 45 °C, where the coulombic efficiency exhibits a rapid decline. The decrease in coulombic efficiency indicates reduced reversibility of lithium-ion migration from the anode during discharge, leading to a large amount of lithium deposition on the anode surface and loss of activity. Therefore, the rapid decrease in coulombic efficiency is a significant cause of the rapid discharge capacity decay.

3.1.2. Charge-discharge curves and midpoint voltage

Fig. 4(a) shows the voltage-capacity curves of the ternary battery cycling at 25 °C. With an increase in the cycle number, the charging

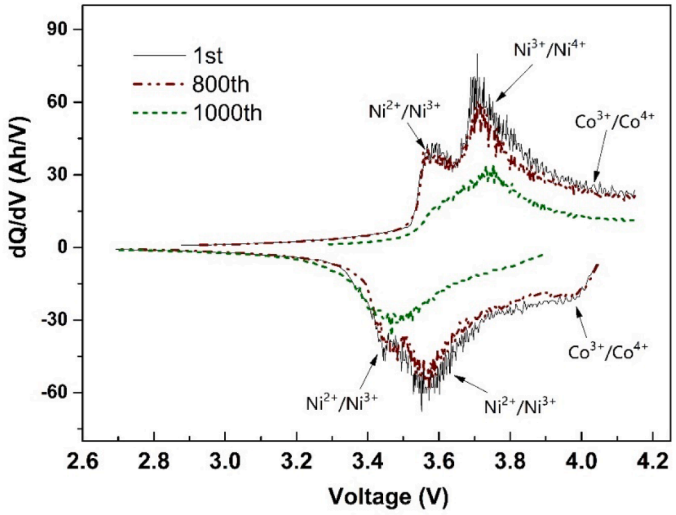


Fig. 6. Capacity differential curves of the battery cycling at 45 °C.

voltage plateau slightly increases, while the discharging voltage plateau shows a slight decrease, indicating increased polarization due to an increase in internal resistance during battery aging. Notably, around the 500th cycle, performance optimization was observed compared to the initial cycle. This is because during the initial charge-discharge cycles, the formation effect of internal materials outweighed the deterioration, but as the formation process completed, the subsequent material deterioration dominates, leading to a decline in battery performance.

Fig. 4(b) shows the variation curve of the midpoint voltage with the number of cycles during the battery cycling test. The midpoint voltage initially shows an upward trend during the first 100 cycles, indicating continuous activation and gradual performance improvement during this stage. However, a significant decrease is observed after 1500 cycles, indicating a rapid deterioration of the battery's performance.

Fig. 5 shows voltage-capacity and voltage data at the higher temperature (45 °C). From Fig. 5(a), it can be observed that during the first 800 cycles, the charge-discharge curves of the battery overlap well in the early stages, with only minor differences observed at the end of discharge and charge phases. This indicates that the polarization increase in the battery during the first 800 cycles is not significant, and the shortening of the charge-discharge plateau is the direct reason for the decrease in battery capacity, suggesting that the capacity decline is mainly caused by the loss of active materials. However, within the 800–1000 cycle range, the charging plateau of the battery significantly rises, the discharging plateau substantially decreases, and the discharge capacity also decreases significantly. This indicates a different failure mechanism during this interval compared to the first 800 cycles. From Fig. 5(b), it can be observed that the midpoint voltage of the battery at 45 °C is very stable during the first 800 cycles. It can be seen that high-temperature cycling is more beneficial for the activation of the battery's performance. However, between cycles 800 and 900, the discharge midpoint voltage starts to show a rapid drop, which corresponds to the decrease in the charge-discharge plateau and the capacity decline.

To further study the phenomenon of rapid drop in capacity at the end of the high-temperature cycling life of the ternary battery, the capacity differential processing of the charge-discharge curves during high-temperature cycling is performed. Fig. 6 shows the capacity differential curves of the ternary battery cycling at 45 °C for the initial cycle, 800 cycles, and 1000 cycles. It can be seen that there is little change in the potential of each oxidation–reduction peak during the initial cycle and 800 cycles, indicating that the potential polarization increase during this process is relatively small and the potential is stable. During the charging process, oxidation peaks of Ni^{2+}/Ni^{3+}, Ni^{3+}/Ni^{4+}, and Co^{3+}/Co^{4+} appear at 3.58 V, 3.72 V, and 4.1 V, respectively. During the discharging process, reduction peaks of Co^{3+}/Co^{4+}, Ni^{3+}/Ni^{4+}, and Ni^{2+}/Ni^{3+} appear near 3.97 V, 3.58 V, and 3.45 V, respectively [17]. The polarization voltage of the electrode is about 0 .13V.

At the 1000th cycle, the oxidation–reduction peaks decrease sharply, indicating the manifestation of capacity degradation. The oxidation peak potential of Ni^{3+}/Ni^{4+} slightly increases to 3.75 V, while the reduction peak during discharge decreases to 3.45 V, resulting in an increase in battery polarization by 0.3 V, which is one reason for the capacity decay. It is worth noting that the weakening of the

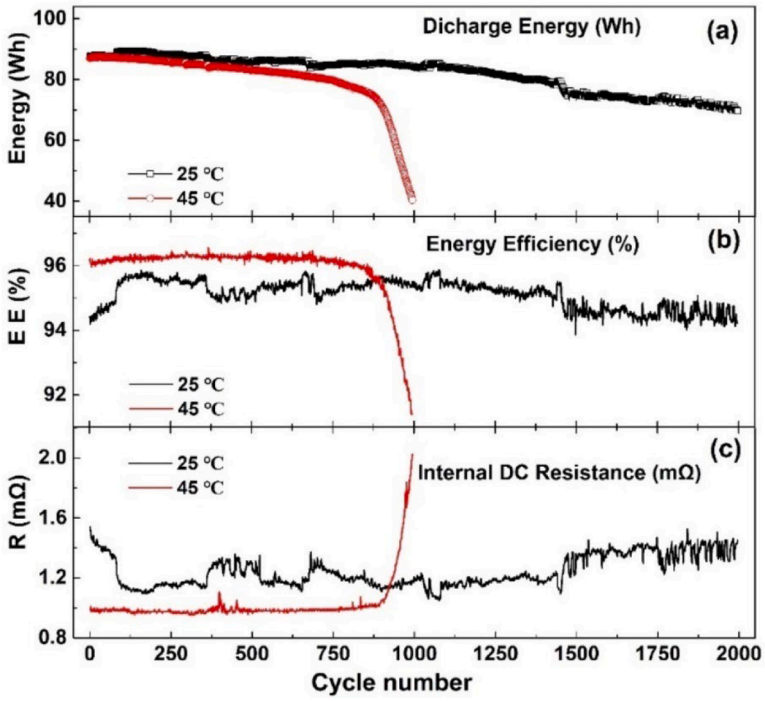


Fig. 7. Discharge energy (a), discharge energy efficiency (EE) (b) and full state DC internal resistance (c) versus cycle number at 45 °C and 25 °C.

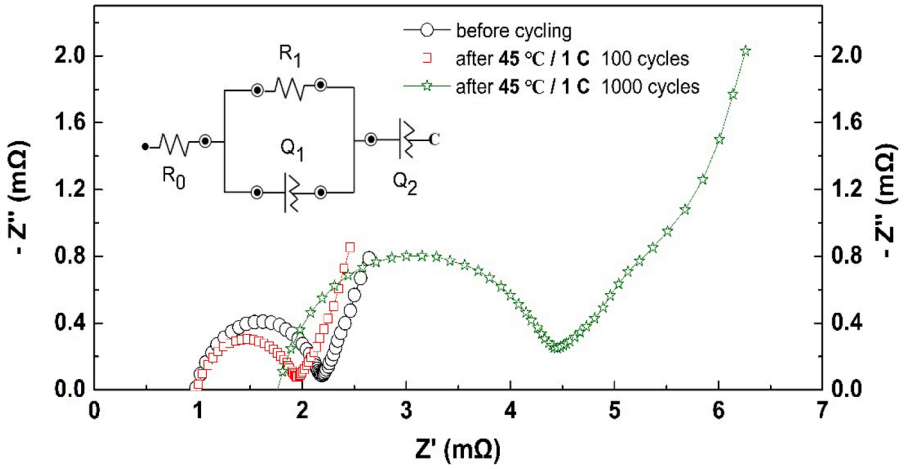


Fig. 8. The evolution of impedance during cycling at 45 °C with 1C current and Nyquist plots of battery with the equivalent circuit.

oxidation–reduction peak of Ni^{2+}/Ni^{3+} is more pronounced, indicating severe loss of Ni^{2+}. This may be because some Ni^{2+} ions enter the lithium layer and lose activity, also exacerbating the degree of lithium-nickel atom mixing in the crystal. Lithium-nickel atom mixing is one of the reasons for the performance degradation of ternary lithium-ion batteries [18].

3.1.3. Discharge energy, energy efficiency, and full charge DC internal resistance

During battery discharge, not all the charged energy can be fully converted into electrical energy, as electrochemical reactions always involve energy losses by battery polarization and ohmic resistance. This part of the energy is mainly released in the form of heat. From Fig. 7(a), it can be observed that the discharge energy variation during battery cycling follows a similar pattern to the discharge capacity variation, but with some differences. This is mainly because capacity is the integral of current over time, while energy is the integral of power over time. High-temperature (45 °C) cycling shows a rapid decline in discharge energy after 800 cycles, while cycling at 25 °C exhibits a slow decrease in discharge energy.

Fig. 7(b) shows the discharging energy efficiency during cycling. The initial energy efficiency of the 25 °C cycling is 94.3%, which continuously increases during the subsequent 100 cycles, reaching a maximum of 95.8%. As the cycle number increases, the energy efficiency of the battery gradually decreases, indicating that the internal active materials of the battery experience an activation-deactivation process. In the case of high-temperature (45 °C) cycling, the discharge energy efficiency stabilizes at 96.1% for the first 800 cycles but exhibits a rapid decline afterward.

In this study the DC internal resistance was calculated by $R_{DC} = \Delta U / \Delta I$ where ΔU and ΔI are change in the voltage and current of the battery in the first 10 s after it discharges from the full charged state. R_{DC} here includes ohmic resistance and polarization resistance, and it is commonly used to evaluate battery health and predict its lifespan in practical applications. During 25 °C cycling, the DC internal resistance of the battery first decreases and then increases, while in the case of 45 °C cycling, the DC internal resistance remains stable for the first 800 cycles and is lower than that observed during 25 °C cycling. This suggests that at 45 °C, lithium-ion migration is faster, and electrochemical reactions are more stable. However, after 900 cycles, there is a turning point with a subsequent increase in DC internal resistance. The start of gassing, the rapid increase in DC internal resistance and the rapid decrease in battery discharge energy almost occur simultaneously. It is likely that due to the continuous accumulation of failure factors, the internal resistance of the battery increases, leading to an increase in heat production and exacerbating the gassing of the battery. Comparing

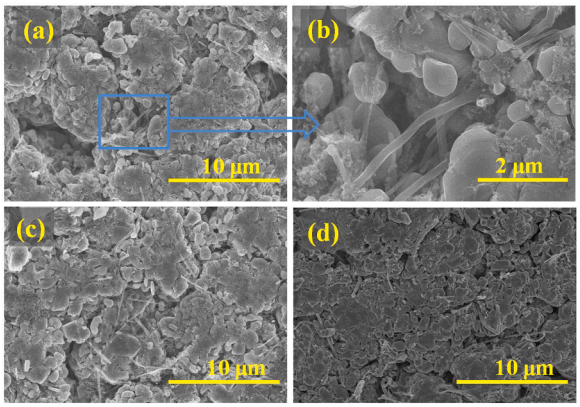


Fig. 9. SEM images of different cathodes: (a) and (b) cathode of a pristine battery at different magnifications; (c) cathode after 2000 cycles at 25 °C; (d) cathode after 1000 cycles at 45 °C.

Fig. 7(b) and (c), it can be observed that the energy efficiency variation is exactly opposite to the trend of DC internal resistance, suggesting that the energy loss during discharge is mainly due to internal resistance consumption.

Fig. 8 shows the impedance results of the battery with different cycles at 45 °C with 1C current. The results are shown as Nyquist plots with an equivalent circuit. R_0 represents the Ohmic resistance, which is the sum of the migration resistance of lithium ions in the electrolyte, the contact resistance between battery components, the current collector resistance, and the electrode resistance. R_1 reflects the charge transfer resistance and lithium ion migration resistance at the electrode interface. The R_0 values of the new battery, after 100 cycles and 500 cycles are 0.97 mΩ, 0.99 mΩ, and 1.76 mΩ, respectively. The R_1 values are 2.19 mΩ, 1.96 mΩ, and 4.43 mΩ, respectively. After the battery 100 cycles, R_0 increases and R_1 decreases. This indicates that during this process, on one hand, the ohmic resistance slightly increases, on the other hand, the ability of the battery to conduct charges/ions is enhanced due to high-temperature activation. But after 1000 cycles, the battery had gassed severely, leading to significant increase in both R_1 and R_0 of the battery. This indicates that gassing is a key factor in the increased impedance during later stages of high-temperature cycling. As impedance rises, heat generation accelerates, further intensifying gassing and creating a feedback loop that worsens battery performance.

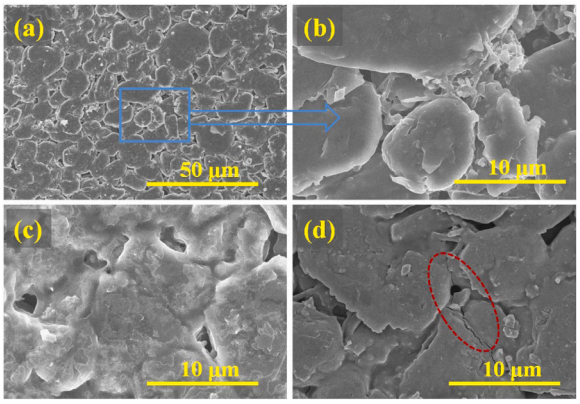


Fig. 10. SEM images of different anodes: (a) and (b) anode of a new battery at different magnifications; (c) anode after 2000 cycles at 25 °C; (d) anode after 1000 cycles at 45 °C.

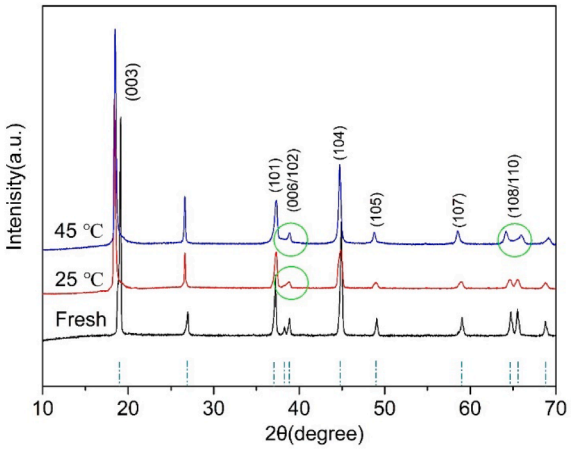


Fig. 11. XRD patterns of the NCM cathode for the pristine battery, and the batteries after 2000 cycles at 25 °C and 1000 cycles at 45 °C.

3.2. Component failure of batteries during normal and high-temperature cycling

To study the factors that trigger high-temperature failure in ternary batteries, fully discharged batteries cycled at both 25 °C and 45 °C were disassembled inside a glove box to analyze the physicochemical properties of the cathodes and anodes.

3.2.1. Electrode microstructure

Fig. 9 shows SEM images of the surface of the cathode at different magnifications for a pristine battery (a) and (b), the cathode after 2000 cycles at 25 °C (c), and the cathode after 1000 cycles at 45 °C (d). The ternary cathode material consists of primary NCM particles in the range of 0.2–1 μm, forming larger aggregates of 3–10 μm in size. This design mainly aims to improve the battery's electrical conductivity and compactness. The conductive agent and carbon nanotubes are evenly dispersed, jointly enhancing the material's electrical conductivity. After 2000 cycles at 25 °C, no significant changes are observed on the electrode surface, but after 1000 cycles at 45 °C, the battery particles agglomerate together, and the boundaries of the material particles become unclear. This indicates irreversible changes occurring on the cathode side during the high-temperature cycling.

Fig. 10 shows SEM images of the anode surface at different magnifications for a pristine battery (a) and (b), the anode after 2000 cycles at 25 °C (c) and the anode after 1000 cycles at 45 °C (d). The graphite anode's particle size is in the range of 5–20 μm, and the pristine battery's

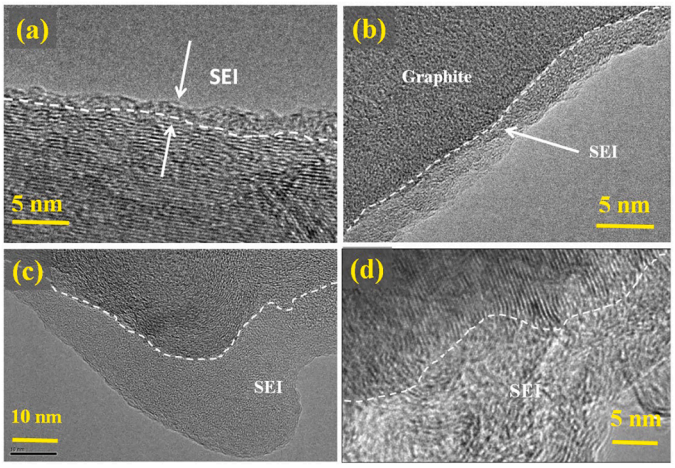


Fig. 12. TEM images of the anode of batteries: (a) pristine; (b) 2000 cycles at 25 °C; (c) and (d) 1000 cycles at 45 °C.

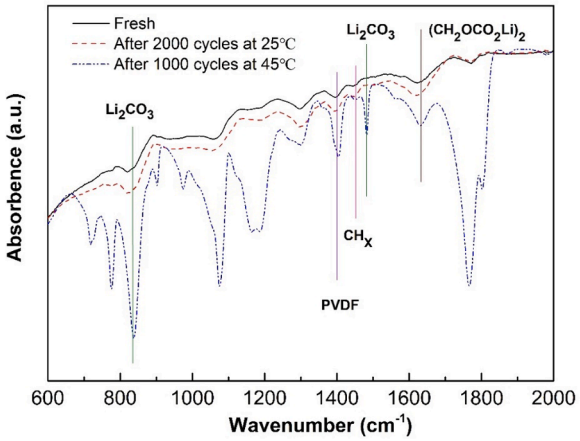


Fig. 13. IR of the anode of batteries: pristine, after 2000 cycles at 25 °C and 1000 cycles at 45 °C.

graphite has a clear and transparent boundary, which is the SEI formed on the graphite surface during the formation process. After 2000 cycles at 25 °C, the SEI on the graphite surface thickens and covers the surface of the graphite particles. After 1000 cycles at 45 °C, "dust-like" deposits appear on the graphite surface, accompanied by a fracture appearance. This may be due to gas generation in the battery leading to the evaporation of the electrolyte and the settling of insoluble substances on the graphite surface.

To explore the changes in the crystal structure of the cathode at different temperatures during cycling, the dismantled NCM cathode was analyzed by XRD, and the results are shown in Fig. 11. The degree of splitting of two sets of peaks, (006/102) and (108/110), is an important indicator of the ordered structure of the NCM layered-structure [19]. It can be seen that the pristine battery's NCM exhibits a well-crystallized layered crystal structure. After 2000 cycles at 25 °C, overlapping occurs between the (006) peak and the (101) peak, indicating that the layered-structure is somewhat affected. After 1000 cycles at 45 °C, the (006) peak and the (108) peak shift to lower angles, and the (006) peak and the (101) peak overlap. The distance between the (108) peak and the (110) peak increases, indicating a change in the NCM layered-structure.

3.2.2. Analysis of anode interface of batteries after normal and high-temperature cycling

Fig. 12 shows TEM images of the anode surface of a battery before

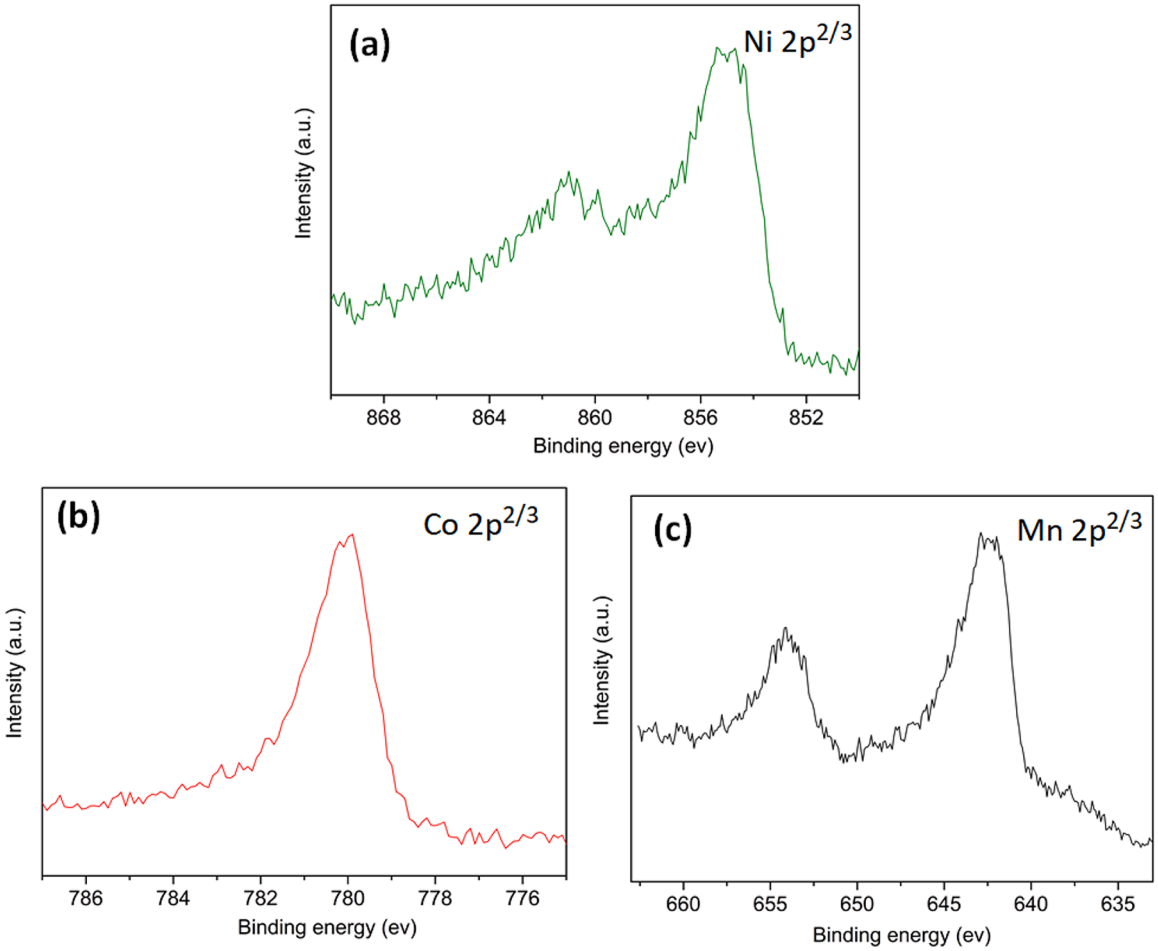


Fig. 14. XPS spectra of transition metal elements at the surface of the anode after 1000 cycles at 45 °C: (a) nickel element absorption spectrum; (b) cobalt element absorption spectrum; (c) manganese element absorption spectrum.

(a), and after 2000 cycles at 25 °C (b), and after 1000 cycles at 45 °C (c, d). A 2–3 nm thick SEI is observed on the surface of the pristine battery, which thickens to 3–5 nm after 2000 cycles at room temperature. After 1000 cycles at 45 °C, the SEI at the anode interface thickens further to 10–20 nm. The SEI mainly consists of lithium-containing carbon-hydrogen compounds. The thickening of the SEI leads to a loss of active lithium in the battery system and an increase in internal resistance, accelerating battery performance degradation.

To study the chemical environment and composition changes at the anode interface during battery failure, infrared analysis was performed on the anode after different cycling conditions. Fig. 13 shows the infrared (IR) spectra of the anode after a new battery, after 2000 cycles at 25 °C, and after 1000 cycles at 45 °C. The absorption peaks at 1490 cm^{-1} and 835 cm^{-1} correspond to characteristic peaks of Li_2CO_3. The absorption peak at 1400 cm^{-1} is characteristic of PVDF. The absorption peak at 1488 cm^{-1} corresponds to C—H compounds, and the peak at 1634 cm^{-1} corresponds to $(CH_2OCO_2Li)_2$ [20-22]. Li_2CO_3, C—H compounds, and RO-Li are the main components at the anode interface and the major constituents of the SEI. After cycling at room temperature, the anode interface shows little change, but after cycling at 45 °C, the Li_2CO_3 absorption peak becomes significantly enhanced. This may be due to gas generation during high-temperature cycling, leading to a large amount of Li_2CO_3 deposition on the anode surface and contributing to the thickening of the SEI.

3.2.3. *Failure mechanism of high-temperature cycling ternary batteries*

Energy dispersive spectroscopy (EDS) was initially used to analyze the anode after cycling at 45 °C. However, it was not sensitive enough to detect nickel, cobalt, and manganese. Therefore, a combination of XPS, ICP, and AES was chosen to study the deposition of transition metal elements on the anode, providing both qualitative and quantitative analysis. Fig. 14 shows the XPS spectra of transition metal ions on the surface of the anode after 1000 cycles at 45 °C. The spectra indicate the presence of nickel, cobalt, and manganese elements near the surface of the anode (0–3 nm). It can be inferred that these transition metal atoms come from the dissolution of cathode materials. Based on the binding energy positions of nickel, cobalt, and manganese, it is determined that these elements exist in the oxidized states of Ni^{2+}/Ni^{3+}, Co^{2+}/Co^{3+}, and Mn^{4+} rather than in their metallic forms [23–26].

Table 1
Metal element content (excluding collector) in the anode of pristine battery, after 2000 cycles at 25 °C, and after 1000 cycles at 45 °C.

Transition metal element content (ppm)	Ni	Co	Mn
Pristine battery	7	4	15
25 °C cycle (2000 weeks)	39	26	41
45 °C cycle (1000 weeks)	55	49	94

ICP-AES was used to detect the elemental content of the anode powder (excluding the aluminum foil collector). Table 1 shows the mass percentage of metal elements in the anode of a new battery, after 2000 cycles at 25 °C, and after 1000 cycles at 45 °C. All anodes of all batteries contain nickel, cobalt, and manganese transition metal elements. The content of transition metal elements in the anode of the new battery is relatively low, which may be due to a small amount of transition metal

Table 2
Gas composition percentage (%) after 1000 cycles at 45 °C.

Gas composition	CO_2	H_2	CO	CH_4	C_2H_4	C_2H_6	C_3H_6	C_3H_8
Percentage	52.24	21.99	12.58	4.11	2.21	2.48	0.62	3.77

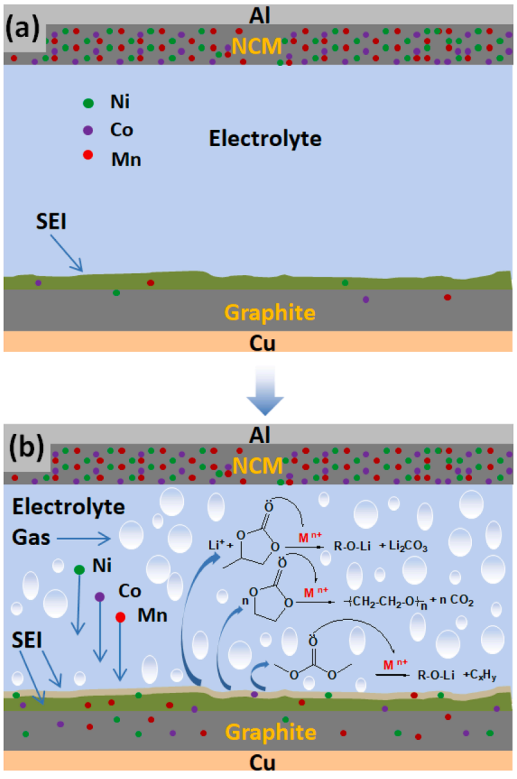


Fig. 15. Schematic illustration of the failure mechanism of the ternary battery during 45 °C cycling: (a) first stage of no-gassing; (b) second stage of deposition of transition metals, thickening of SEI, and gas generation during rapid battery performance degradation.

dissolution during the formation and storage process of the battery and their deposition on the anode. After 2000 cycles at 25 °C, the content of transition metal elements in the anode increases significantly compared to the new battery, and after 1000 cycles at 45 °C, the deposition of transition metal elements on the anode becomes more severe, with the manganese element reaching as high as 94 ppm.

Table 2 shows the gas composition in a ternary battery after 1000 cycles at 45 °C. The main gas components are CO_2, H_2, and CO, with CO_2 content as high as 52.24%. Other gas components include C1-C3 small alkanes, with a total content proportion of 13.19%. The presence of C_3H_6 and C_3H_8 in the gas indicates decomposition of propylene carbonate (PC) in the electrolyte solvent [27].

3.3. *Mechanism of battery failure*

With all the results presented above, especially the data for the ternary lithium-ion batteries subjected to long-term cycling at 25 °C and 45 °C, a possible mechanism for the failure of the batteries in the late stage of the high-temperature cycling can be proposed, as depicted in Fig. 15. Fig. 15(a) shows the first stage of the cycles absent of gassing. There is very little transition metal on the anode surface, and the SEI film is clearly visible. During high-temperature cycling, the transition metal ions in the cathode of the battery continuously dissolve in the electrolyte, migrate over the electrolyte/separator towards anode (driven by the electrical field during the charging steps), and deposit on the anode, as show in Fig. 15(b). The deposited transition metal atoms have strong catalytic activity and can combine with the lone pair electrons of the C = O bond in organic solvents, causing ring opening and chain breaking reactions in cyclic organic solvents such as propylene carbonate and ethylene carbonate, and chain breaking reactions in chain solvents such as dimethyl carbonate.

In the later stage, the transition metal elements continuously precipitate at the anode, catalyzing the gas generation reaction in organic solvents. This leads to the deposition of $LiCO_3$ and insoluble RO-Li on the anode surface, causing a continuous thickening of the SEI. This process leads to an increase in the electrochemical resistance and ohmic resistance of the battery, resulting in a continuous increase in the heat generation of the battery, which further exacerbates gas generation behavior. Thus, gas generation, increased resistance, and higher battery temperature are mutually coupled, leading to the phenomenon of rapid drop in battery performance at high temperature.

4. Conclusions

The entire life cycles of commercial pouch ternary battery operated at 25 °C and 45 °C can be divided into two stages. The first stage is characterized by a slow performance decline, mainly due to minor changes in the cathode layered-structure, thickening of the anode SEI film, and a slow increase in the internal resistance of the battery. The second stage is marked by a rapid performance decline, particularly evident at the higher temperature of 45 °C showing a rapid deterioration in the performance. This second stage differs in failure mechanisms from the first stage of battery degradation. During high-temperature cycling at 45 °C, the battery's discharge capacity, coulombic efficiency, median discharge voltage, discharge energy, and energy efficiency decline rapidly, and the direct current internal resistance rises sharply. The coupling of gas generation, increased resistance, and higher battery temperature drives the rapid performance degradation in the later stage of the ternary battery's lifespan. At the anode interface, gas generation mainly involving CO_2 occurs, accompanied by the deposition of Li_2CO_3 and thickening of the SEI film. The deposition of transition metal ions at the anode interface catalyzes the gas generation reaction in the electrolyte solvent responsible for rapid drop in the performance of ternary batteries during high-temperature cycling. Measures such as coating or reconstructing a stable cathode electrolyte interphase (CEI) film on the surface of the cathode material to reduce the dissolution of transition metal ions may be an effective way to inhibit gas generation during high-temperature cycling and prolong the lifespan of ternary batteries.

CRediT authorship contribution statement

Suijun Wang: Writing – original draft, Methodology. **Jialiang Liu:** Resources, Methodology. **Jerry Y.S. Lin:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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021).

Data availability

Data will be made available on request.

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